Depth First Search (DFS)

C Program Implementation and Sample Output

PROGRAM SOURCE CODE

```
#include <stdio.h>
#define MAX 50

int graph[MAX][MAX];
int visited[MAX];
int n;

void DFS(int vertex)
{
    int i;
    // Mark vertex as visited
    visited[vertex] = 1;
    // Display vertex
    printf("%d ", vertex);
    // Visit adjacent vertices
    for (i = 0; i < n; i++)
    {
        if (graph[vertex][i] == 1 && visited[i] == 0)
        {
            DFS(i);
        }
    }
}

int main()
{
    int i, j;
    int start;
    // Read number of vertices
    printf("Enter number of vertices: ");
    scanf("%d", &n);
    // Read adjacency matrix
    printf("Enter the adjacency matrix:\n");
    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &graph[i][j]);
        }
    }
    // Initialize visited array
    for (i = 0; i < n; i++)
    {
        visited[i] = 0;
    }
    // Read starting vertex
    printf("Enter starting vertex: ");
    scanf("%d", &start);
    printf("DFS Traversal: ");
    DFS(start);
    return 0;
}
```

SAMPLE INPUT & EXECUTION OUTPUT

```
Enter number of vertices: 5
Enter the adjacency matrix:
0 1 1 1 0
1 0 1 0 0
1 1 0 0 1
1 0 0 0 0
0 0 1 0 0
Enter starting vertex: 0

DFS Traversal: 0 1 2 4 3
```

Note: DFS output can change depending on the order in which adjacent vertices are checked. In this program, vertices are checked from 0 to n-1.